

SAI MOUNIKA PEDAMALLU

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SUMMARY

Motivated and detail-oriented Health informatics graduate with a strong foundation in healthcare data analysis, information management, and technology integration. Skilled in electronic health records, data interoperability, and information security. Excels in leveraging emerging technologies to enhance patient care and streamline healthcare processes. Committed to continuous learning and professional growth in the rapidly evolving field of Health Informatics.

EDUCATION

Masters in Health Informatics

September 2023 - May 2025

HOFSTRA UNIVERSITY | Hempstead, New York

Course work: Data Visualization, Advanced Data Analytics, Clinical Decision Support Systems, AI &ML/ Robotics for Medical Informatics, Database Management System, Security and privacy in Healthcare, Emerging Technologies IOT.

Bachelor of Pharmacy

September 2019 - May 2023

Narasaraopeta Institute of Pharmaceutical Sciences, | Narasaraopet, India

PROFESSIONAL EXPERIENCE

Graduate Research Project

January 2025 - May 2025

Hofstra University | Hempstead

- Developed a Clinical Decision Support Dashboard as part of the Acute Stroke Mortality Prediction Project, leveraging SPARCS data (2013–Present) to improve clinician insights into stroke outcomes.
- Built machine learning models in Python (Scikit-learn, Pandas) to predict in-hospital and 30-day stroke mortality, incorporating SHAP for model interpretability.
- Designed an interactive dashboard UI featuring real-time vitals, lab result tracking, sepsis criteria checklist, and evidence-based treatment suggestions, using CSS Flexbox/Grid.
- Integrated Flask-based APIs to connect model outputs with front-end visualizations, enabling live updates of risk scores and alerts.
- Optimized backend architecture for scalable deployment and ensured HIPAA-compliant handling of clinical data.
- Applied Agile methodology through weekly sprints, retrospectives, and collaborative iteration cycles with data scientists and frontend developers.

Healthcare Business Analyst

September 2024 - December 2024

Spring Broad Incubators | New York

- Designed and implemented a Health-Related Social Needs (HRSN) screening application for identifying barriers such as housing insecurity, food insecurity, transportation needs, and financial strain using the AHC CMS screening tool.
- Collaborated with interdisciplinary teams to define screening criteria, incorporating 13 core domains including safety, mental health, disabilities, and family/community support.
- Built backend workflows to process responses through pre-defined truth tables and determine eligibility for referral to Community-Based Organizations (CBOs).
- Enabled seamless patient navigation through a QR code-based system, linking users to personalized CBO resources upon screening completion.
- Conducted backend testing and process flow validation to ensure accurate referral logic and improve community health engagement.

- Led mock SCN digital health implementation and presented solution design to stakeholders, aligning with CMS and public health standards.

Internship

June 2024 - August 2024

HITLAB | New York, NY

- Planned a hypothetical digital healthcare charity event at the InterContinental New York Barclay venue, expecting 500 attendees, organized using Excel and Zoom platforms
- Collaborated with renowned speakers in the tech and healthcare sectors to drive a 20% increase in event attendance and engagement
- Developed a comprehensive event budget of \$50,000 utilizing Excel functions such as Pivot Tables, VLOOKUP, and SUMIF to ensure financial feasibility
- Boosted event visibility by 100% through spearheading the creation of outreach materials, such as flyers, aligning marketing efforts, and securing commitments
- Proposed and implemented a strategic data-driven proposal to the HITLAB team, resulting in a 20% increase in attendee engagement and a 15% rise in event conversion rates

PHARMACISTS

January 2023 - July 2023

Diagnostic Labs | India

- Introduced barcode scanning and automated inventory systems, reducing supply chain costs by 20% and improving process efficiency.
- Delivered motivational counseling and health education, contributing to a 40% increase in patient satisfaction and adherence.
- Conducted FDA-aligned workshops on GMP, quality control, and regulatory compliance, driving a 30% rise in employee engagement.
- Ensured strict compliance with FDA protocols in production and retail environments, leading to a 20% improvement in product quality and staff adherence.

PROJECTS AND RESEARCH

Organization: Hofstra University

MedFlow: AI-Powered Optimization of Inpatient Pharmacy Workflows.

Health Innovation Project: Developed AI mattress solutions for managing bedsores, integrating cost-benefit analyses and business strategies.

Data Analysis Project: Utilized Tableau and SQL for healthcare quality analysis, exploring trends in maternal mortality and social determinants of health.

Cybersecurity in Healthcare: Researched emerging technologies and frameworks to enhance data protection in healthcare systems.

Creating descriptive & predictive analytics in healthcare by choosing specific KPI's : LOS, Readmission Rate, Mortality Rate, Bed Utilization Rate, ICU stay Duration from mimiciii Database. Languages used during the project include Python, SQL, VSCODE, Microsoft Excel, Tableau.

STRENGTHS

- Troubleshooting and Problem-solving skills
- Strong knowledge in Pharmacological and medical terminology
- Strong organizational and time management skills
- Exceptional communication and interpersonal skills
- Ability to work independently and as part of a team

- Detail-oriented and able to handle multiple tasks simultaneously

Technical SKILLS

Python, SQL, Tableau, Data Analysis, Data Mining, Quantitative Analysis, Statistical Modeling, Predictive Modeling, Data Manipulation, Power BI, SPSS, Hypothesis Testing, Complex Data Sets, Data Visualization Techniques, MS Office.

Honors & Achievements

- A Review article on Qualification of Anti Coagulants - Dabigatran, Rivaroxaban by chromatographic & Spectrometric techniques.
- Development of UV- Spectrophotometric Method for the Determination of Sildenafil and Fluoxetine in Bulk & Formulation.